Wipro — Aptitude Test 4

Questions 76–100 of 100 · 25 questions

Question 76. A starts 3min after B for a place 4.5 km distant. B on reaching his destination immediately returns and after walking 1 km meets A. If A can walk 1km in 18min, then what is B's speed?

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Solution:
Assume that the time taken by B to walk
4.5 + 1 = 5.5 km is = T
Time taken for A to meet B = T – 3.
The distance travelled by A to meet B
= 3.5 km.
Time taken for A to cover 3.5 km
= 3.5 x 18
= 63 minutes
= 1 hour and 3 minutes.
= 1.05 hours
Time taken by B to cover 5.5km
= 63 + 3
= 66 minutes
= 1.1 hours
Speed of B = 5.5/1.1
= 5 kmph

Question 77. A race course is 400m long. A and B run a race and A wins by 5m. B and C run over the same course and B win by 4m. C and D run over it and D wins by 16m. If A and D run over it, then who would win and by how much?

Solution:
Speed of train = 120/2.5 = 48m/s
Time taken to cross a platform
T = (Lt + Lp)/St
Lt = length of train
Lp = length of platform
St = Speed of train
T = (120 + 140)/48
T = 5.41 seconds.

Solution:
Ratio between distance of A and B when
A finishes the race = 400 : 395
Ratio between distance of B and C when
B finishes the race = 400 : 396
Ratio between distance of C and D when
D finishes the race = 384 : 400.
A : D = (A/B)(B/C)(C/D)
A : D = (400/395)(400/396)(384/400)

A : D = (400 x 384)/(395 x 396)
A : D = 2560 : 2607
A/D = 2560/2607
D will beat A. the distance covered by A
when D covers 400 m will be:
A/400 = 2560/2607
A = 2560 x 400/2607
A = 392.8m
D beats A by 7.2m.

Question 78. A train travelled from A to B and back in a certain time at rate of 60kmph. If the train had travelled from A to B at a rate of 80kmph and back from B to A at a rate of 40kmph it would have taken 2 hrs longer. Find the distance between A and B.

W i p r o

Solution:
Assume that the distance between
A and B = D.
Assume that the time taken at 60kmph
to travel from A to B and B to A = 2T
Time taken in the second case = 2T + 2.
From the first case:
D = 2T x 60 = 120T
Average speed in the second case:
= 2(80x40)/(80+40) = 53.33
From second case:
D = 53.33 x (2T – 2) = 106.67T+106.67
120T = 106.67T + 106.67
T = 8 hours.
Distance between A and B
= 8 x 60 = 480km.

Question 79. 120m long train crosses the pole in 2.5 seconds. Find how much time will it take to cross a 140m long platform?

Question 80. Two men together start a journey in the same direction. They travel at a speed of 9 kmph and 15 kmph respectively. After travelling for 6 hours the man travelling at 9kmph doubles his speed and both of them finish the

distance in the same time. How many hours will they take to reach their destination?

Solution:
The distance covered by first person in 6 hours = 6 x 9 = 54 km
The distance covered by second person in 6 hours = 6 x 15 = 90km
Distance between the two persons at this point is = 90 – 54 = 36km.
Time taken for the first person to catch the second person after doubling his speed:
T = 36/(12 – 9) = 36/3 = 12 hours.
Total tie of the journey
= 6 + 12 = 18 hours.

Question 81. The ratio between speeds of 2 trains is 7:8. If the 2nd train runs 400km in 4 hours what is the speed of 1st train?

Solution:
Speed of second train = 400 km in 4hours
= 400/4 kmph
= 100kmph.
Speed of 1st train:
7/8 = x/100
Speed of first train = 87.5 kmph

Question 82. A car travelling at 5/7th of its actual speed covers 42km in 1hr 40min 48 sec. What is the actual speed of the car?

Solution:
Write 1 hour 40 minutes 48 seconds in decimal value.
40 minutes = 40 x 60 = 2400 seconds
40 minutes 48 seconds
= 2400 + 48 = 2448 seconds
= 2448/3600 = 0.68 hours
So, 1 hour 40 minutes 48 seconds
= 1.68 hours.
Speed of the car at reduced speed
= 42/1.68
= 25kmph
Actual speed of the car
= 25 x 7/5 = 35kmph.

Question 83. A man can row a distance of 5 km in 60 min with the help of the tide. The direction of the tide reverses with the same speed. Now he travels a further 20 km in 10 hours. How much time he

would have saved if the direction of tide has not changed?

Solution:
Speed of the boat when the tide is on the same direction = 5kmph
To cover 5 + 20 = 25km
The time taken will be 25/5 = 5 hours
Actual time taken was
1 + 10 = 11 hours.
If the tide has not changed its direction, he would have saved
11 – 5 = 6 hours.

Question 84. When not moving on the sidewalk, Maya can walk the length of the sidewalk in 7 minutes. If she stands on the sidewalk as it moves, she can travel the length in 4 minutes. If Maya walks on the sidewalk as it moves, how many minutes will it take her to travel the same distance? Assume she always walks at the same speed, and express your answer as a decimal to the nearest tenth.

Solution:
Assume that the speed length of the side walk is 280 m. (assuming a value which is a common multiple of 4 and 7)
Speed of Maya while walking near the sidewalk
= 280/7 = 40 meter per minute
Speed of the sidewalk
= 280/4 = 70 meter per minute.
Speed of Maya when walking on the side walk = 40 + 70 = 110 m/minute
Time taken when Maya walks on the sidewalk
= 280/110
= 2.54 minutes
= 2.5 minutes (nearest tenth decimal).

Question 85. Maria drove to the mountains last weekend. There was heavy traffic on the way there, and the trip took 9 hours. When Maria drove home, there was no traffic and the trip only took 4 hours. If her average rate was 40 miles per hour faster on the trip home, how far away does Maria live from the mountains?

Solution:
Assume that the distance from home to mountain = D
Assume that her average speed on the way to mountain = S kmph
So, her speed down the mountain = S + 40 kmph.
Calculating distance while going up the mountain:
D = 9S
Calculating distance while coming down the mountain:
D = 4(S + 40)
Therefore from the above two equations, we get
9S = 4S + 160
5S = 160 ■S = 32kmph.
Distance from home to mountain = Speed x time taken.
Considering the equation
D = S x 9
We get,
= 32 x 9
= 288km

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Question 86. A train leaves New York City at 7.15 am and arrives in Buffalo at 2.47 that afternoon. What total length of time does the trip take?

Solution:
An example of a 7 digit number with the digit 3 three times and the digit 5 five time is:
3335555
Number of ways of arranging
= 7!/(3!x4!)
= 35

Solution:
Time taken for the journey
= 02:47am – 07:15am
Time from 07:15 to 12:00 = 4 hours 45 minutes.
Time from 12:00 to 2:47 = 2 hours 47 minutes.
Total time = 04:45 + 02:47

= 07:32 hours
7 hours 32 minutes.

Question 87. Two trains travel in the same direction with 56 kmph speed. The first train crosses a person standing in 78.9sec. Find the length of the first train.

Solution:
The first word in the alphabetical order will be: “EHMORT”
Assign number for each letter in ascending order
E = 1; H = 2; M = 3; O = 4; R = 5; T = 6
The word written in terms of number = 346215
There are totally 6! = 720 words possible.
Among the numbers 346215, ‘3’ comes third in ascending order.
So, (3 -1)5! = 240
Among the numbers 46215, ‘4’ comes third in ascending order.
So, (3 -1)4! = 48
Among the numbers 6215, ‘6’ comes 4th

Solution:
Distance travelled by the train to cross the man = length of the train.
Length of the train = Speed x time
Speed of the train
= 56 x 5/18 = 15.55 mps
Length of the train
= 15.55 x 78.9
= 1226.8 m

Question 88. Given 10 letters out of which 5 are to be chosen. How many words can be made with at least one repetition.

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in ascending order.
So, (4-1)3! = 18
Among the numbers 215, ‘2’ comes 2nd

Solution:
Number of ways of arranging with at
least
1
repetition
=
Number
of
arrangements with repetition – Number
of arrangements without repetition.
Number of arrangements with repetition
= 105

in ascending order.
So, (2-1)2! = 2
By arranging the remaining numbers 15
in ascending order rank of 15 is 1.
Adding all the values:
240 + 48 + 18 + 2 + 1
= 309.
Rank of the word Mother = 309

= 100000
Number
of
arrangements
without
repetition
= 10P5 = 10x9x8x7x6 = 30240
Number of arrangements with at least
one repetition
= 100000 – 30240
= 69760

Question 89. How many 7-digit numbers are formed having digit 3 three times and digit 5 four times?

Question 90. What is the rank of the word "MOTHER“ when all the letters of the word are arranged in dictionary order?

Question 91. In a game each person is dealt three cards from a deck of 52 cards & a player is said to have a winning deck if & only if he or she has a king, queen & a jack each

irrespective of the color of the sign.
What is the total possible number of winning decks for this game?

Solution:
Number of ways of having 1 king = 4C1 = 4 ways
Number of ways of having 1 queen = 4C1 = 4 ways
Number of ways of having 1 jack = 4C1 = 4 ways
Total number of ways of having a winning deck = 4 x 4 x 4 = 64 ways.

Question 92. There are 30 socks in a drawer. 60% of the socks are red and the rest are blue. What is the minimum number of socks that must be taken from the drawer without looking in order to be certain that at least two blue socks have been chosen?

Solution:
Number of red socks = (60/100)30 = 18
Number of blue socks = 30 – 24 = 12
By drawing the socks the first 18 socks may be red. While drawing the 19th sock we can be sure that we have got 1 blue sock.
by drawing the 20th sock we can be sure that we have 2 blue socks.
The minimum number of socks to be drawn = 20.

Question 93. Mr. Varma has 4 different paintings that he wishes to divide among his 3 children. In how many ways can he do this if each child must get at least 1 painting?

Solution:
Selecting 3 paintings for three persons can be done in 4C3 = 4 ways.
The tree paintings can be arranged among the three children in 3! = 6 ways.
The remaining 1 painting can be given to any one of the children in 3C1 = 3 ways.
Total ways of giving the paintings = 4 x 6 x 3 = 72.

Question 94. A, B and C are three speakers. They have to speak randomly along with 5 other speakers in a function. What is the probability that A speaks before B and B speaks before C?

Solution:
Among the three speakers A, B and C they can be arranged in 3! = 6 ways.
Out of the six ways in only 1 way A will be before B and B will be before C.
In all the arrangements 1 out 6 ways A will be before B and B will be before C.
The probability = 1/6

Question 95. If A and B are 2 numbers and are selected randomly from the values 1, 2, 3, 4, 5, … 25 what is the probability that A and B are not equal?

Solution:
The number of ways in which two numbers can be selected fro the numbers 1 to 25 = 25C2 = 300
The selections in which the numbers are same = (1,1), (2,2), (3,3), … (25,25)
There are 25 selections in which the two numbers will be same.
The number of selections in which the two numbers will not be same
= 300 – 25 = 275
Probability = 275/300 = 11/12

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Question 96. A and B each throw a dice. What is the probability that A's throw is not greater than B?

Solution:
Total number of results when two dice are thrown = 62 = 36
The number of results in which the two results will be same = 6
Remaining results = 30.
Number of results in which A will be less than B = 30/2 = 15
It is said that A’s throw is not greater than B’s throw.
A’s throw is less than or equal to B’s throw
Probability = (15 + 6)/36 = 21/36

Question 97. There are 3 red balls, 3 green balls and 3 blue balls. What is the probability that when three balls are selected at least two of them are of same color?

Solution:
The time between noon and midnight = 12 hours.
The number of times the two hands overlap is asked between 12pm and 12 am. So those two times cannot be taken into account.
The hour hand and minute hand overlaps between (1 and 2), (2 and 3), (3 and 4), … and (10 and 11).
Number of times the two hands overlap = 10 times.

Solution:
Total ways of selecting 3 balls with at least two balls of same color
= Total number of selections – number of selections in which all balls are of different colors.
Total ways of selecting three balls = 9C3 = 84 ways
To have 3 balls from different colors, we have to select 1 from each color.
Number of ways of selecting three different colors = 3C1 x 3C1 x 3C1 = 27
Number of selections in which there are at least two balls of different colors
= 84 – 27 = 67
Probability = 67/84

Question 98. In an objective exam which has 2 answer options each for all the 20 questions, what is the probability that a person answers all the questions correctly? (Assume that he answers all the questions)

Solution:
Number of ways of answering all the questions = 220

Number of ways of answering all the questions correctly = 120 = 1

Probability = 1/220

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Question 99. Two dice are tossed. What is the probability of getting an even number on first die or sum of the results is 8?

Solution:
Total number of results = 62 = 36
Number of results in which the sum is 8 = 5, {(2,6),(3,5),(4,4),(5,3),(6,2)}
The number of results in which the first dice will have even number = 36/2 = 18
The 18 results in which the first dice has even number, 3 results are common to the results having sum equal to 8.
Probability = (5+18 – 3)/36 = 20/36

Wipro was the first

company to
implement the Six
Sigma. The company

is very particular
about the quality and

the process.
Wipro currently has

close to 160,000
employees across the

globe. The company
has employees of 98
nationalities from 61
different countries in

the world. The
presence of Wipro can

be seen in more than
175 cities and 8.5% of

the work force of
Wipro is non-Indian.

Question 100. In a conventional clock, how many times does the minutes hand pass the hour hand between noon and midnight?